
Lecture Notes

on Hardy Weinberg Equilibrium

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1 Alleles and their frequencies

Consider a large population of organisms with a single gene that has two alleles:

A (dominant allele), a (recessive allele).

Let

$$p = P(A) \quad \text{and} \quad q = P(a)$$

denote the *allele frequencies* of A and a in the population.

Since A and a are the only alleles at this locus and are mutually exclusive events, by the axioms of probability,

$$P(A) + P(a) = 1 \Rightarrow p + q = 1.$$

$$\begin{aligned} p &= 0.3 \\ q &= ? \\ 1 - 0.3 & \end{aligned}$$

2 Genotype frequencies under random mating

We now find the expected genotype frequencies in the next generation under **random mating**.

The possible genotypes from these two alleles are

AA , Aa , aa .

	A	a
A	AA	Aa
a	aA	aa

Under random mating, each offspring receives one allele from each parent, and each allele is chosen **independently** with probabilities p and q .

- Probability of genotype AA :

$\hookrightarrow A \hookrightarrow a$

$$P(AA) = P(A \text{ from one parent}) \cdot P(A \text{ from the other parent}) = p \cdot p = p^2.$$

- Probability of genotype Aa :

There are two ways to obtain Aa : A from the first parent and a from the second, or a from the first parent and A from the second.

$$P(Aa) = P(A \text{ then } a) + P(a \text{ then } A) = pq + qp = 2pq.$$

$$p a = a p$$

$Aa \quad aA$

- Probability of genotype aa :

$$P(aa) = P(a \text{ from one parent}) \cdot P(a \text{ from the other parent}) = q \cdot q = q^2.$$

$$ax^2 + bx + c = 0.$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Since AA , Aa and aa are mutually exclusive and exhaustive genotypes, their probabilities must sum to one:

$$P(AA) + P(Aa) + P(aa) = 1.$$

Substituting the expressions above,

$$p^2 + 2pq + q^2 = 1,$$

which is simply

$$(p + q)^2 = 1,$$

consistent with $p + q = 1$.

$$\Rightarrow p + q = 1$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$0 \leq P(E) \leq 1.$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

2.1 Punnett square view

The same result can be seen with a Punnett square. Place the alleles from one parent across the top and from the other parent along the side.

	A	a
A	AA	Aa
a	Aa	aa

If the allele frequencies in the parents are p for A and q for a , the four boxes occur with probabilities

$$p^2, \quad pq, \quad qp, \quad q^2,$$

which again give genotype frequencies p^2 for AA , $2pq$ for Aa , and q^2 for aa .

2.2 Statement of the Hardy Weinberg Equilibrium

Under the following conditions

- very large population size,
- random mating,
- no selection,
- no mutation,
- no migration,

the allele frequencies p and q remain constant from generation to generation, and the genotype frequencies in the population are

$$P(AA) = p^2, \quad P(Aa) = 2pq, \quad P(aa) = q^2.$$

The phenomenon that the allele and genotype frequencies remain the same from one generation to the next under these assumptions is called the **Hardy Weinberg equilibrium**.

Example. In a population of field mice living in a large grassland, fur color is determined by a single gene with two alleles:

$\overset{p}{A}$ (dominant, gray fur) and $\overset{q}{a}$ (recessive, white fur).

The population is assumed to be in Hardy Weinberg equilibrium. A recent genetic survey found that the frequency of homozygous dominant individuals (AA) in the population is 36%. If the total population of field mice is 800, answer the following questions.

1. What is the frequency of allele A in the population?

$$AA = p^2$$

2. How many mice in the population are expected to have genotype AA ?

$$Aa = 2pq$$

3. How many mice are expected to have genotypes Aa and aa ?

$$aa = q^2$$

Let $P(A) = p$ and $P(a) = q$.

$$\text{Given, } P(AA) = p^2 = 0.36 \Rightarrow p = \sqrt{0.36} = 0.6$$

$$\therefore q = 1 - p = 1 - 0.6 = 0.4.$$

$$\begin{cases} 100 \\ 20\% \\ \rightarrow 100 \times 20\% \\ = 20 \end{cases}$$

$$1) P(A) = p = 0.6.$$

2) The number of mice in the population with genotype AA is $P(AA) \times 800 = 0.36 \times 800 = 288$.

$$3) P(Aa) = 2pq = 2 \times 0.6 \times 0.4 = 0.48.$$

$\therefore$ The number of mice in the population with genotype Aa is $P(Aa) \times 800 = 0.48 \times 800 = 384$.

$$P(aa) = q^2 = (0.4)^2 = 0.16.$$

$\therefore$ The number of mice in the population with genotype aa is $P(aa) \times 800 = 0.16 \times 800 = 128$.

☑ **Example.** Assume Hardy–Weinberg equilibrium for the recessive genetic condition *moro*. The fraction of the population that has *moro* is 0.20.

1. What fraction of the population are *carriers* of *moro*?
2. Suppose a person has *moro* and one of their parents is known to have *moro*. What is the probability that the other parent also has *moro*?
3. One parent is known to be a carrier, and the second parent's genotype is known to be either *Mm* or *MM*. If they already have one child *without* *moro*, what is the probability that their second child will have *moro*?

Let M be the normal allele and m be the recessive allele that causes the condition *moro*.

Let $P(M) = p$ and $P(m) = q$. $\therefore p + q = 1$.

Now under the Hardy Weinberg equilibrium, the genotype frequencies are

$$P(MM) = p^2, \quad P(Mm) = 2pq \text{ and } P(mm) = q^2.$$

Given, $q^2 = 0.20 \Rightarrow q = \sqrt{0.20} \approx 0.4472$.

$$\therefore p = 1 - q = 1 - 0.4472 = 0.5528.$$

a) $P(Mm) = 2(0.5528)(0.4472) \approx 0.494 \approx 49.4\%$.

b) Let H_1 = denote the 2nd parent having genotype *mm*.

1st parent
mm

2nd parent
 $\swarrow \searrow$
mm *Mm*

H_2 = denote 1st " " " "
genotype *Mm*.

	<i>m</i>	<i>M</i>
<i>m</i>	<i>mm</i>	<i>Mm</i>
<i>M</i>	<i>Mm</i>	<i>MM</i>

$$\therefore P(H_1) = P(mm) = q^2 = 0.20, \text{ and } P(H_2) = P(Mm) = 2pq = 0.494.$$

let C denotes the event that the child has more (mm).

$$g: P(H_1 | C) = ?$$

$$P(C|H_1) = ?$$

$$P(C|H_2) = ?$$

$$P(H_1) = v$$

$$P(H_2) = \checkmark$$

$$P(C | H_1) = P(\text{child mm} | H_1)$$

1st parents already having mm

	m	m'
m	mm	mm'
m'	mm	mm'

$$= \frac{4}{4} = 1.$$

$$P(c | H_2) = P(\text{child with mm} | H_2)$$

$$= \frac{2}{4} = \frac{1}{2}$$

	m	m
M	Mm	Mm
m	mm	mm

Now by Bay's theorem,

$$P(H_1 | c) = \frac{P(c | H_1) P(H_1)}{P(c)}$$

$$= \frac{P(C|H_1) P(H_1)}{P(C|H_1) P(H_1) + P(C|H_2) P(H_2)}$$

$$= \frac{(1 \times 0.20)}{(1 \times 0.20) + (\frac{1}{2} \times 0.494)}$$

$$\approx 0.4474.$$

$$\begin{aligned}
 &= \frac{1 \cdot a^2}{(1 \cdot a^2) + \cancel{\left(\frac{1}{2} \times 2a\right)}} \\
 &= \frac{a^2}{a^2 + 1a} \\
 &= \frac{a^2}{\cancel{a}(a+1)} \\
 &= \frac{a}{1} \\
 &= a
 \end{aligned}$$